

Melt Onset Variability in the Northern Canadian Arctic Archipelago From Synthetic Aperture Radar and Passive Microwave Satellite Observations, 1979–2024

Claire Driedger , Mallik Mahmud , Maisha Mahboob , Stephen E.L. Howell , Alex Cabaj, and Mike Brady

Abstract—The timing of sea ice melt onset (MO) in the Arctic is important because it controls the surface energy balance and influences the amount of sea ice that survives the melt season. Here, we investigate the timing of sea ice MO in the Queen Elizabeth Islands (QEI), located within the Last Ice Area of the Canadian Arctic Archipelago, using Sentinel-1 synthetic aperture radar (SAR) imagery from 2015 to 2024. The Last Ice Area is the most stable ice region in the Arctic, and changes within this region provide insight into the overall health of Arctic sea ice. We used a threshold-based approach to detect MO for every SAR pixel using time-series backscatter. We contextualized the MO record within the framework of the earlier SAR record and the long-term passive microwave (PMW) record. The QEI region exhibited considerable interannual variability in the timing of MO, as detected by SAR, with a mean MO date occurring on day of year 157 (± 6 days). From 2015 to 2024, no significant trend toward earlier MO was observed; instead, SAR and PMW data showed a slight delay. Extending the analysis back to 1997, utilizing RADARSAT-1/2 data revealed a 5-day per decade earlier MO trend, whereas PMW data (1979 – 2022) indicated a 1.31 to 1.49-day per decade decrease. Sentinel-1 SAR provided superior spatial detail relative to PMW, particularly near coastlines and in narrow channels and showed greater sensitivity to melt detection, with discrepancies in trends attributed to methodological and data variations.

Index Terms—Arctic, C-band, melt onset (MO), sea ice, sentinel-1.

I. INTRODUCTION

SURFACE air temperatures across the Arctic have been increasing at a rate that is three to four times faster than the global mean [1], [2], and this increase has been attributed to human influence [3]. The timing of melt onset (MO) over Arctic sea ice is triggered by enhanced downwelling longwave

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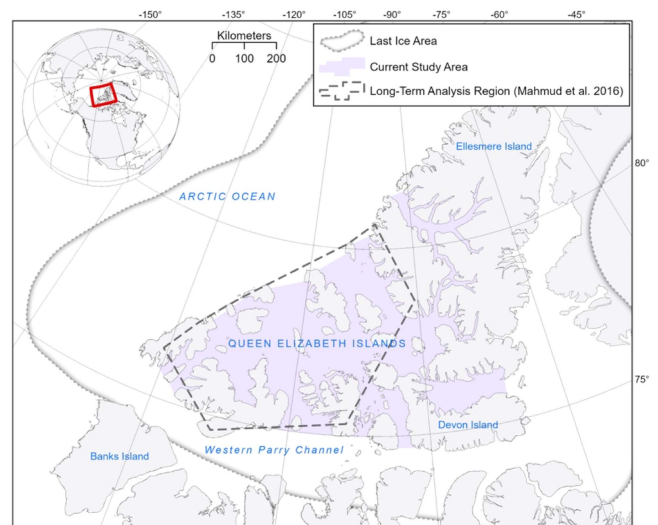


Fig. 1. Study area location of the northern CAA, also referred to as the QEIs, where we estimate MO timing. The dashed region is the location of the previous study area used in [15].

radiation [4], but melt also coincides with high solar radiation [5], [6]. The timing of MO controls the Arctic surface energy balance, as exemplified by the sea ice-albedo feedback that influences the total heat input into the ocean during the summer season and, in turn, the amount of sea ice that survives the melt season [5]. Decreasing trends in Arctic sea ice are associated with an earlier timing of MO over the last several decades [7], [8], [9]. The corresponding heat input into the ocean can also cause delays in fall freeze-up [10], [11], [12] and suppress ice growth during winter [13].

The Last Ice Area refers to the region surrounding the northern Canadian Arctic Archipelago (CAA) and Greenland (see Fig. 1). While the Arctic Ocean is projected to become seasonally sea ice-free during the summer months (i.e., sea ice extent < 1 million km^2), sea ice in the Last Ice Area is expected to remain [14]. This region will act as a last refuge for various marine-dependent species. Here, we focus on the region of the Last Ice Area located within the CAA, collectively referred to as the Queen Elizabeth Islands (QEI), building on a previous study [15] from 1997 to 2014. Given its relatively short melt season, the timing of

MO impacts sea ice processes within the QEI. Sea ice within the QEI is primarily composed of multi-year ice (MYI) that is landfast for most of the year (November to July) [16], [17]. Sea ice thickness within the QEI ranges from 3.4 to 5.5 m, with minimal evidence of thinning since the 1970s [16], [18]. Sea ice area within the QEI has also remained relatively stable over the past 50+ years as it continues to receive a steady supply of thick MYI from the Arctic Ocean during the melt season [19], [20]. Sea ice within the QEI is exported southward to the shipping channels of the Northwest Passage during the melt season and represents a considerable hazard to navigation [21]. Continued monitoring of sea ice processes within this region is of considerable importance, and monitoring its stability can provide insight into the overall health of Arctic sea ice.

Detecting the timing of MO from satellite-based microwave remote sensing has a long history, as it does not depend on sunlight availability and is not hindered by cloud coverage. Methods that use passive microwave (PMW) observations collected by the Scanning Multichannel Microwave Radiometer (SMMR), the Special Sensor Microwave Imager, the Special Sensor Microwave Imager and Sounder, Advanced Microwave Scanning Radiometer and Advanced Microwave Scanning Radiometer 2 provide the most spatiotemporally consistent estimates of MO and provide an opportunity to assess long-term variability but suffer from a relatively coarse spatial resolution of ~ 12 to 25 km [9], [22], [23], [24], [25]. Active microwave scatterometers such as the Advanced Scatterometer (ASCAT) or the Quick Scatterometer (QuikSCAT), which utilize resolution-enhanced data scatterometer image reconstruction techniques [26], [27] have also been widely utilized for detecting the timing of MO [4], [28]. The resolution-enhanced scatterometer datasets provide higher spatial resolution (~ 2.2 to 4.5 km) but have a shorter temporal domain compared to PMW sensors. Synthetic aperture radar (SAR) satellites such as RADARSAT-1/2, which provide high spatial resolution (50–100 m), have also been used to detect MO, but the application has primarily been regionally focused because of a lack of image availability and access [15], [29], [30], [31]. Moreover, SAR satellites lack the long data continuity of PMW or spaceborne scatterometers, resulting in comparatively shorter SAR MO dataset records. With the launch of Sentinel-1 in 2014, the opportunity to detect MO across larger regions or extend the monitoring record of existing datasets is now possible.

Here, we use Sentinel-1 to detect the timing of MO within the QEI region of the northern CAA. To our knowledge, there have been no high-resolution, long-term estimates of MO over the northern CAA (i.e., QEI) since 2014. First, we describe the approach that builds on [15] and apply the algorithm to the QEI region from 2015–2024. We then discuss the corresponding spatial and temporal variability from 2015–2024. Finally, we place the 2015–2024 period within the context of the earlier record (1997–2014) provided in [15], and the long-term record (1979–2022) provided by PMW from [24].

II. DATA

The primary dataset used in this analysis is Sentinel-1 SAR imagery in extra-wide at HH polarization mode from all ascend-

ing and descending orbits from 1 April (late winter) to 30 June (early summer) during 2015 to 2024 over the QEI region of the northern CAA. The extra-wide mode acquires imagery with a 410 km swath width, an incidence angle range of 18.9° – 47.0° , a spatial resolution of 90 m, and a pixel spacing of 40 m. Consistent with other MO monitoring work (see [32]), the use of the HH co-polarization channel throughout the methodology optimizes MO detection because horizontally polarized channels are more sensitive to changes in snow conditions during melt compared to vertically polarized channels. The imagery is available Online.¹

For ice type identification, we make use of the regional weekly sea ice charts provided by the Canadian Ice Service (CIS) to determine sea ice stage of development, partial concentration and coverage within the QEI [17]. The CIS charts are a compilation of in-situ, satellite and aerial reconnaissance data and include information on ice concentration and stage of development, with the primary data source being RADARSAT since 1996. A thorough description of the charts is provided in [33]. The CIS ice charts are available weekly for regions including the Western and Eastern Canadian Arctic from 1968 to present Online.²

We also use the MO estimated by the PMW algorithm described in [24] that is available across the Arctic from 1979. Briefly, this method uses multiple indicators as each is sensitive to different features of melt. The MO signal is determined by summing up the normalized magnitude of each indicator and the day with the largest sum is used. Accordingly, the dataset provides both the first day of melt (i.e., PMW-EMO) and the period of continuous melt (i.e., PMW-CMO). The data is available from NASA's Cryosphere Science Research Portal.³

To provide contextual parameters during the melt season, we use 2-m air temperature (SAT), albedo (α), and mean surface downward shortwave ($K_\downarrow$) and longwave ($L_\downarrow$) radiation flux over the months of April to June from 2015 to 2024 from the ERA5 fifth generation reanalysis dataset by the European Centre for Medium-Range Weather Forecasts [34]. Data is available Online.⁴

III. METHODOLOGY

The methodology used in this study is built on [15]. Nevertheless, all statistical data is derived from newly generated datasets and subsequently compared with literature.

A. SAR Image Processing

We use Google Earth Engine to process raw level-1 ground range-detected Sentinel-1 images by removing border noise and thermal noise (post-July 2015), and then calibrated them to the backscatter coefficient sigma naught (σ^0) [35]. The images are then orthorectified using ASTER DEM.⁵ Using a log scale,

¹[Online]. Available at: <https://dataspace.copernicus.eu/>.

²[Online]. Available at: <https://www.Canada.ca/en/environment-climate-change/services/ice-forecasts-observations/latest-conditions/archive-overview.html>.

³[Online]. Available at: <https://earth.gsfc.nasa.gov/cryo/data/arctic-sea-ice-melt>.

⁴[Online]. Available at: <https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5>.

⁵[Online]. Available at: <https://asterweb.jpl.nasa.gov/gdem.asp>.

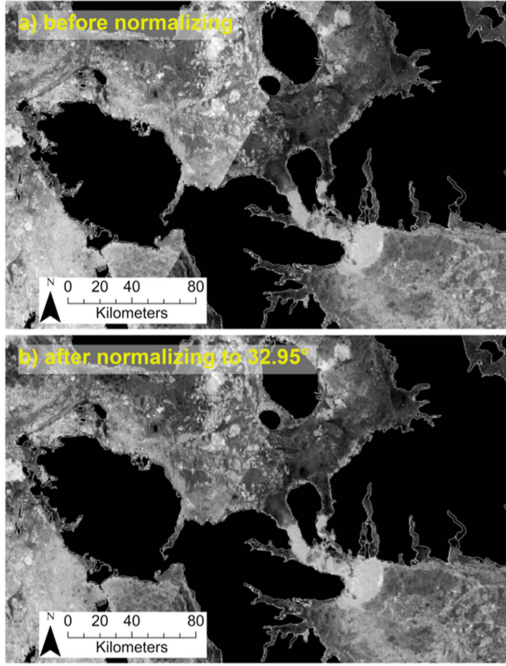


Fig. 2. Sentinel-1 backscatter intensities (a) before and (b) after incidence angle normalization. The image was normalized to 32.95° using the frequency-specific dependence of -0.19 dB/1° . Note the sharp changes in backscatter at the extremities of the swath zone, which correspond to the extremities of the incidence angle range. Sentinel-1 SAR mosaic imagery was acquired on 5 April 2021.

the unitless pixel values are converted to dB units. Only HH polarization was used in this study to remain consistent with [15].

B. Incidence Angle Normalization

The value of σ^0 depends on frequency, polarization, and incidence angle (θ), and the physical characteristics of the sea ice and snow cover, surface roughness including brine content, air bubble inclusions, and snow thickness and grain size [30]. To detect environmentally-driven changes in σ^0 coincident with MO, it is first necessary to apply an incidence angle correction algorithm across the image [36]. Following [37], we use a simple pixel-by-pixel linear scaling to normalize the backscatter ($\sigma^0_{\text{normalized}}$) as follows:

$$\sigma^0_{\text{normalized}} = \sigma^0 - \theta_d (\theta - \theta_{\text{mid}}) \quad (1)$$

where θ_d is the incidence angle dependence, θ is the original incidence angle, and θ_{mid} is the incidence angle of the center of the image. We use an incidence angle dependence of -0.19 dB/1° [37] and normalize every scene to the center incidence angle of 32.95° . An example region of the QEI before and after incidence angle normalization is illustrated in Fig. 2.

C. Sample Site Selection

Thirty first-year ice (FYI) and 30 MYI test sites, for a total of 60 sites, are selected in the QEI for the random sample year of 2021 (see Fig. 3). Site selection is performed using a Sentinel-1 SAR image in April, and CIS ice charts. FYI and MYI types are manually discerned by inspection of the backscatter magnitude

and the textural appearance of the ice. A total of 60 sites are selected to capture the variability of ice types, snow thickness [38], sea ice roughness [39], and sea ice thickness [40], [41] found within the region. Furthermore, sites are selected in areas of 100% ice concentration, and each site is an area of 200 m x 200 m (25 Sentinel-1 pixels), sized to be large enough for statistical significance and small enough for near-homogeneous ice type, roughness, and snow cover pixels. Sites are selected at least 1 km from land to prevent any land contamination in the Sentinel-1 backscatter signal.

D. Sea Ice Type Identification

The large salinity difference between FYI and MYI results in a large backscatter contrast between the ice types, which is maintained throughout the winter months [42]. Indeed, winter-time backscatter is particularly sensitive to both the fractional components of air bubbles and brine inclusions within the upper ice and basal snow layers, as well as to the roughness of the snow-ice interface. These characteristics contribute to the dielectric constant of the snow/ice volume [43], which influences its dominant scattering mechanism and, therefore, the magnitude of backscatter.

For FYI, brine expulsion toward the snow basal layer from the highly saline ice volume inhibits microwave signals from penetrating into the ice volume [44]. Therefore, C-band SAR backscatter is low [45] and dominated by surface scattering, particularly at small θ [37]. For MYI, the radar interacts with the brine-drained and bubbly upper 10–30 cm layer of ice, generating high backscatter throughout the winter months, which is dominated by volume scattering [46]. As a result, it is very important to distinguish between FYI and MYI for representative MO detection.

Accordingly, using all April images over the 30 FYI and 30 MYI test sites, the mean winter backscatter signals (σ^0_w) for FYI ($\sigma^0_{w(\text{FYI})}$) and MYI ($\sigma^0_{w(\text{MYI})}$) are found to be $-23.7 \pm 2.1 \text{ dB}$ and $-10.2 \pm 1.5 \text{ dB}$, respectively. These values are within 3 dB each of the values reported in [15] at an incidence angle of 35° and fall within the FYI and MYI winter backscatter ranges offered by [47]. To discriminate between FYI and MYI, a sea ice type classification threshold ($\sigma^0_{\text{MYI/FYI}}$) is calculated from σ^0_w , following [47]:

$$\sigma^0_{\text{MYI/FYI}} = \sigma^0_{w(\text{MYI})} - 0.4 [\sigma^0_{w(\text{MYI})} - \sigma^0_{w(\text{FYI})}] \quad (2)$$

The sea ice type classification threshold is calculated at $-15.6 \pm 1.8 \text{ dB}$. This agrees with the reported value of -15.7 dB by [15]. It is lower than the MYI/rough FYI classification threshold value of -13.5 dB given in [48], as expected, because smooth FYI generates a low-magnitude backscatter signal, which would decrease the FYI winter mean. As shown in Fig. 4, pixels with $\sigma^0_w < \sigma^0_{\text{MYI/FYI}}$ are classified as FYI, and pixels with $\sigma^0_w > \sigma^0_{\text{MYI/FYI}}$ are classified as MYI.

E. MO Detection

We define MO as the first appearance of liquid water in the snowpack, which is consistent with the established theoretical framework and aligns with previous studies [15], [31], [49], [50],

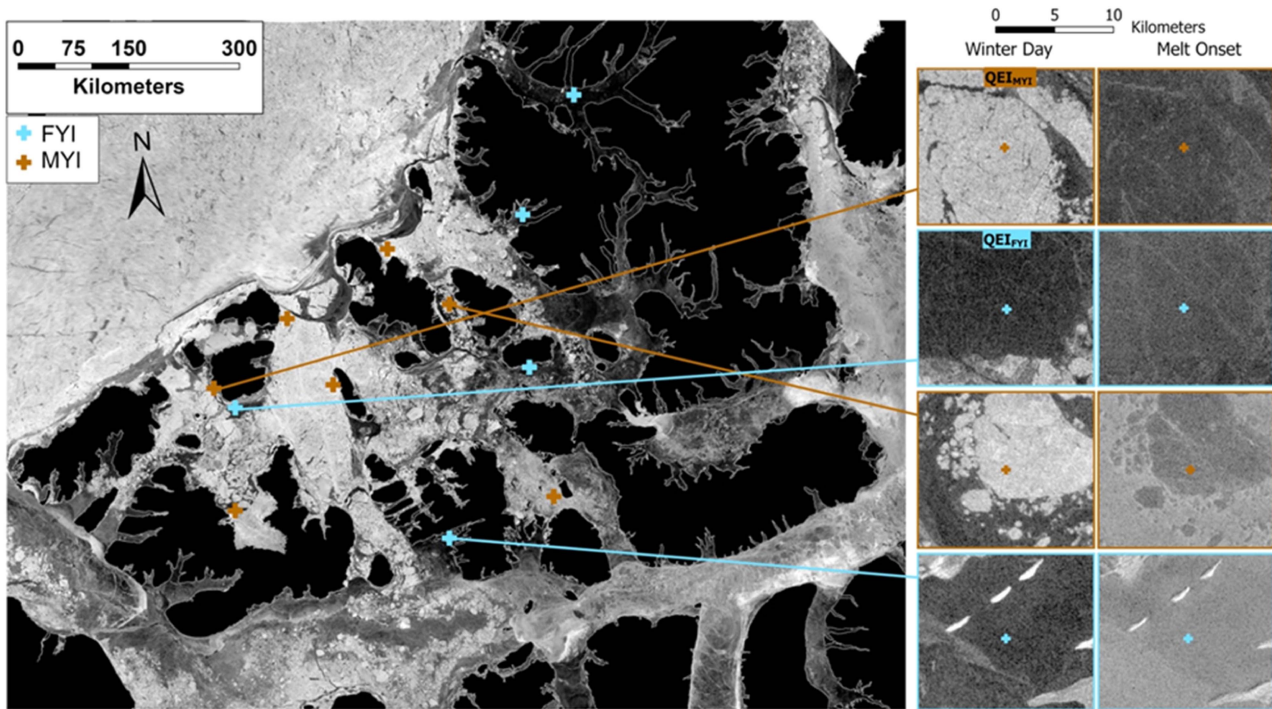


Fig. 3. C-band image composite from Sentinel-1 acquired during 5–7 April 2021. Color-coded validation sites from first-year (FYI) and MYI are highlighted, showing changes in backscatter intensities during winter and MO. Land is shown in black.

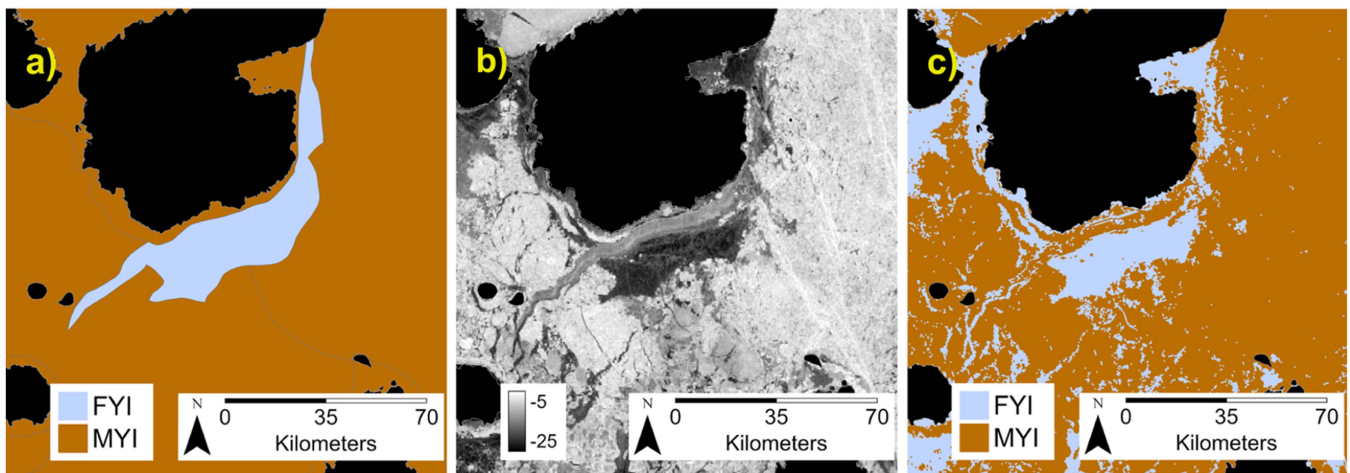


Fig. 4. First-year (blue/dark gray) and multiyear (orange/light gray) ice during winter 2021. (a) Ice types are displayed according to the stage of development of the ice floe (first-year/MYI) with the greatest partial concentration, provided by the CIS (weekly ice chart of 5 April 2021). (b) Sentinel-1 composite image from 5–11 April 2021. The brighter regions correspond to MYI (high backscatter signal) while the darker regions correspond to FYI (low backscatter signal). (c) Results of the ice type classification following [47].

[51], [52] and is similar to the prior definitions [53] of “early melt”. Specifically, at MO, the appearance of liquid water in the snow volume alters the dielectric properties and bulk permittivity of the volume [54], [55], [56], and corresponding changes in backscatter have been quantified for a range of microwave frequencies [30], [46], [53], [57].

Over MYI, a decrease in backscatter is caused by this volumetric moisture, which masks the stronger ice volume scattering [49]. Over FYI, the brine-wetting of enlarged snow grains at the

snow/basal layer (for smooth FYI) [44], [58] and the appearance of meltwater in the snowpack (for rough MYI) introduce a volume scattering component to backscatter, which manifests as an increase in the signal [44], [55], [59], [60].

A transition threshold for detecting MO is developed using a random subset of 12 out of 30 of the test sites, for both ice types. The backscatter on the first day of MO is sampled for the selected sites based on time series analysis. In addition, MO is only triggered if the 1200 (UTC-5) SAT is $> -5^{\circ}\text{C}$, since melt

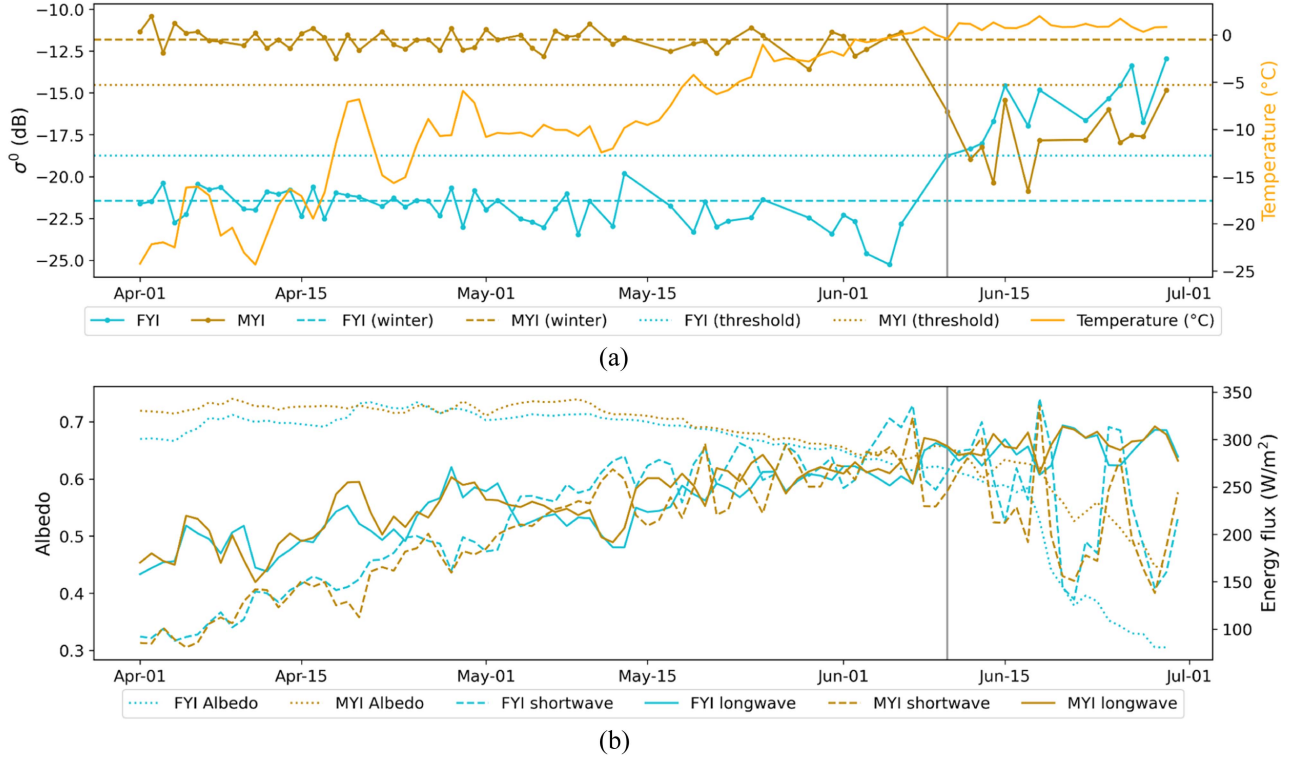


Fig. 5. Temporal evolution of the Sentinel-1 SAR backscatter and near-surface air temperature (a), albedo, downwelling shortwave ($K_{\downarrow}$), and downwelling longwave ($L_{\downarrow}$) energy flux from ERA5 (b) evolution from 1 April to 30 June 2021 for MYI sites (brown) and FYI sites (blue). Color-coded SAR-detected backscatter values are acquired from Sentinel-1, a dotted line indicates the MO threshold, and a dashed line indicates the winter backscatter average (1–30 April). In both panels, gray vertical lines mark the first day when the backscatter exceeds the MO threshold, indicating melt. Melt was detected on 10 June for both ice types.

is not generally found to occur before this point [15], [59], [61]. Following [15], (3) and (4) are used to calculate a transition threshold for both FYI ($\sigma_{\text{TR}/\text{FYI}}^0$) and MYI ($\sigma_{\text{TR}/\text{FYI}}^0$), respectively:

$$\sigma_{\text{TR}(\text{FYI})}^0 = \left(\left(\sigma_{w(\text{FYI})}^0 + \sigma_{w(\text{FYI})(1 \text{ std. dev.})}^0 \right) + \left(\sigma_{\text{MO}(\text{FYI})}^0 - \sigma_{\text{MO}(\text{FYI})(1 \text{ std. dev.})}^0 \right) \right) / 2 \quad (3)$$

$$\sigma_{\text{TR}(\text{MYI})}^0 = \left(\left(\sigma_{w(\text{MYI})}^0 - \sigma_{w(\text{MYI})(1 \text{ std. dev.})}^0 \right) + \left(\sigma_{\text{MO}(\text{MYI})}^0 + \sigma_{\text{MO}(\text{MYI})(1 \text{ std. dev.})}^0 \right) \right) / 2. \quad (4)$$

$\sigma_{w(\text{ice})}^0$ and $\sigma_{w(\text{ice})(1 \text{ std. dev.})}^0$ are the mean and standard deviations for the mean April (winter) backscatter, and $\sigma_{\text{MO}(\text{ice})}^0$ and $\sigma_{\text{MO}(\text{ice})(1 \text{ std. dev.})}^0$ are the mean and standard deviations for the MO backscatter, where “ice” is “FYI” or “MYI” accordingly. To establish the absolute MO thresholds, the difference between the mean winter backscatter and the transition threshold for FYI ($\sigma_{\text{Th}(\text{FYI})}^0$) and MYI ($\sigma_{\text{Th}(\text{MYI})}^0$) is calculated from (5) and (6), following [15]:

$$\sigma_{\text{Th}(\text{FYI})}^0 = \sigma_{\text{TR}(\text{FYI})}^0 - \sigma_{w(\text{FYI})}^0 \quad (5)$$

$$\sigma_{\text{Th}(\text{MYI})}^0 = \sigma_{\text{TR}(\text{MYI})}^0 - \sigma_{w(\text{MYI})}^0. \quad (6)$$

Over the 12 selected sites, the mean MO thresholds for FYI and MYI are 2.71 and -2.71 dB, respectively. Compared to

reported values for FYI and MYI of 2.8 and -3.0 dB, respectively, our transition thresholds will result in at most an early MO detection over FYI of 0.5 days, and no change in MO detection over MYI [15].

Thereafter, $\sigma_{w(\text{ice})}^0$ was calculated for each pixel across the years 2015 to 2024. MO detection was initiated on the first day after May 1 when the daily backscatter surpassed the MO threshold. A detailed description of this algorithm can be found in [15]. Despite the inherent short-term variability of C-band SAR backscatter over sea ice [44], [51], [62], we did not compute a running three-day backscatter average, which is used to prevent early MO detection [15]. Manual inspection of the backscatter time series of all 24 sites used to calculate the transition thresholds revealed that the daily backscatter only exceeded the MO threshold at or after the date when the three-day temporal average also did so, in all cases except one.

IV. RESULTS AND DISCUSSION

A. MO Threshold Evaluation

The temporal evolution of radar backscatter at FYI and MYI sites is linked to environmental factors, including temperature, albedo, and radiative energy, which shape observed backscatter evolution (see Fig. 5). Backscatter over FYI and MYI remains stable in winter, reflecting consistent ice and snow properties. For FYI, surface scattering predominates, while MYI

TABLE 1
SUMMARY OF THE TIMING OF MO DATE, ALBEDO AND SURFACE AIR TEMPERATURE (SAT) ON THE MO DATE (MEAN $\pm$ STANDARD DEVIATION) OVER SEA ICE IN THE QEIS REGION OF THE CAA FROM 2015–2024

Year	MO Date	Albedo	SAT ($^{\circ}$ C)
2015	159 $\pm$ 7	0.54 $\pm$ 0.1	0.6 $\pm$ 1.6
2016	160 $\pm$ 8	0.51 $\pm$ 0.1	1.3 $\pm$ 2.8
2017	155 $\pm$ 12	0.59 $\pm$ 0.1	−0.6 $\pm$ 1.8
2018	159 $\pm$ 5	0.58 $\pm$ 0.1	−0.2 $\pm$ 1.1
2019	158 $\pm$ 5	0.60 $\pm$ 0.1	0.1 $\pm$ 0.9
2020	159 $\pm$ 6	0.60 $\pm$ 0.1	0.4 $\pm$ 0.7
2021	160 $\pm$ 7	0.61 $\pm$ 0.1	−0.6 $\pm$ 0.5
2022	165 $\pm$ 7	0.60 $\pm$ 0.1	0.3 $\pm$ 0.8
2023	158 $\pm$ 8	0.59 $\pm$ 0.1	0.6 $\pm$ 2.2
2024	161 $\pm$ 7	0.54 $\pm$ 0.2	0.8 $\pm$ 1.2
mean	157 $\pm$ 6	0.58 $\pm$ 0.1	0.3 $\pm$ 0.6

backscatter is dominated by volume scattering from bubbles [36], [63]. However, above -5°C , significant diurnal backscatter fluctuations occur. With increasing $K_{\downarrow}$, moisture starts to appear at the snow surface, which results in temporary absorption of radar energy—the classic dip before showing increased backscatter during FYI MO, which can be noticed in Fig. 5. The dielectric constant of snow-covered FYI rises quickly for thinner layers of snow (less than 10 cm) at higher temperatures, as brine and snow grain size increase, enhancing backscatter during MO [64]. In addition, FYI may show a gradual backscatter decrease in winter due to snow accumulation, which typically begins in the first winter after ice formation, reducing snow salinity and scattering properties [30]. In contrast, MYI backscatter varies with snow water content, as liquid water in snow reduces volume scattering from sea ice, leading to an overall decline in scattering during MO compared to winter.

MO was apparent on the first day after both the FYI and MYI daily backscatter crossed the MO threshold (see Fig. 5). For all sites, MO coincided with air temperature rising to near or above freezing and there was a drop in albedo of at least 0.05 between 1 April and MO for all sites (see Table 1). MO was also associated with increasing values of $L_{\downarrow}$ and $K_{\downarrow}$ between 250 and 300 $\text{W}\cdot\text{m}^{-2}$ with $L_{\downarrow}$ being stronger (see Fig. 5). This is consistent with previous studies that have shown that increases in net radiation, primarily associated with increases in downwelling longwave radiation, are influential in triggering the timing of MO [28], [61], [62], [65].

B. Spatial and Temporal Variability of MO From 2015–2024

The time series of the date of Sentinel-1 SAR detected MO as well as both PMW melt indicators (i.e., PMW-EMO and PMW-CMO) is shown in Figs. 6–8 for the current study area. The QEI region experienced considerable interannual variability in the timing of MO as detected by SAR with a mean MO date occurring on day of year (DOY) 157 (± 6 days) that ranged from DOY 155 (± 12) in 2017 to DOY 165 (± 17) in 2022 (see Table 1). Interestingly, there is a slight increasing trend from 2015–2024 observed in all three MO indicators (see Fig. 6). The mean PMW-EMO was DOY 156 (± 5 days), which was similar to the mean MO from SAR over the coincident time period of 2015–2022; however, the correlation was only 0.53 (see Fig. 6).

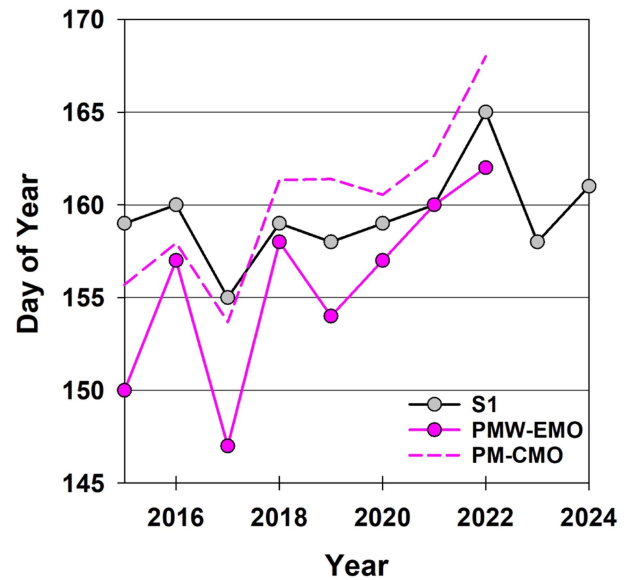


Fig. 6. Time series of the mean date of MO (DOY) as detected by sentinel-1 from 2015–2024 and PMW observations from [24] in the QEIs from 2015–2022.

The mean PMW-CMO DOY was 160 (± 4 days), later than the mean MO from SAR, but the correlation was stronger at 0.61.

The spatial distribution of MO timing as detected by SAR in the QEI from 2015–2024 is shown in Fig. 8. The mean MO timing indicates that MO occurs first in the narrow channels and inlets and progresses spatially from west to east (see Fig. 8). MO was considerably earlier and widespread throughout the QEI in 2017, whereas 2022 MO was considerably later, especially in the eastern regions of the QEI (see Fig. 7). The high level of spatial variability for the SAR MO maps is noteworthy and reflective of the mix of ice types, degree of ridging and inherent snow thickness conditions in the region. This is particularly striking when SAR is spatially compared against both PMW-EMO and PMW-CMO, with the latter not being able to detect MO close to the coastline or in narrow channels and inlets (see Fig. 8).

C. Long-Term Variability in the Timing of MO

Figs. 9 and 10 show the time series of the timing of MO within the region investigated in [15] (see Fig. 1) from 1997 to 2024 (i.e., SAR period) and from the longer PMW record from 1979–2022. Over the period of 2015–2024, Sentinel-1 MO timing exhibited a slight increasing trend (see Fig. 6), but when the Sentinel-1 MO dates from 2015–2024 are combined with the RADARSAT-1/2 MO dates from 1997–2014, the long-term trend indicates an overall decrease in MO [see Fig. 9(a)]. The mean MO date from 1997–2014 was DOY 164 (± 4) and from 2015–2024 was DOY 157 (± 6). Over the entire time period from 1997–2024, the SAR detected MO exhibited a decreasing trend of 5 days decade $^{-1}$ [see Fig. 9(b)]. Over the 1997–2022 period, the trend in PMW-EMO is only -1.60 days decade $^{-1}$ and PMW-CMO is -1.55 days decade $^{-1}$. Over the entire long-term PMW record from 1979–2022, the MO has decreased between 1.31 and

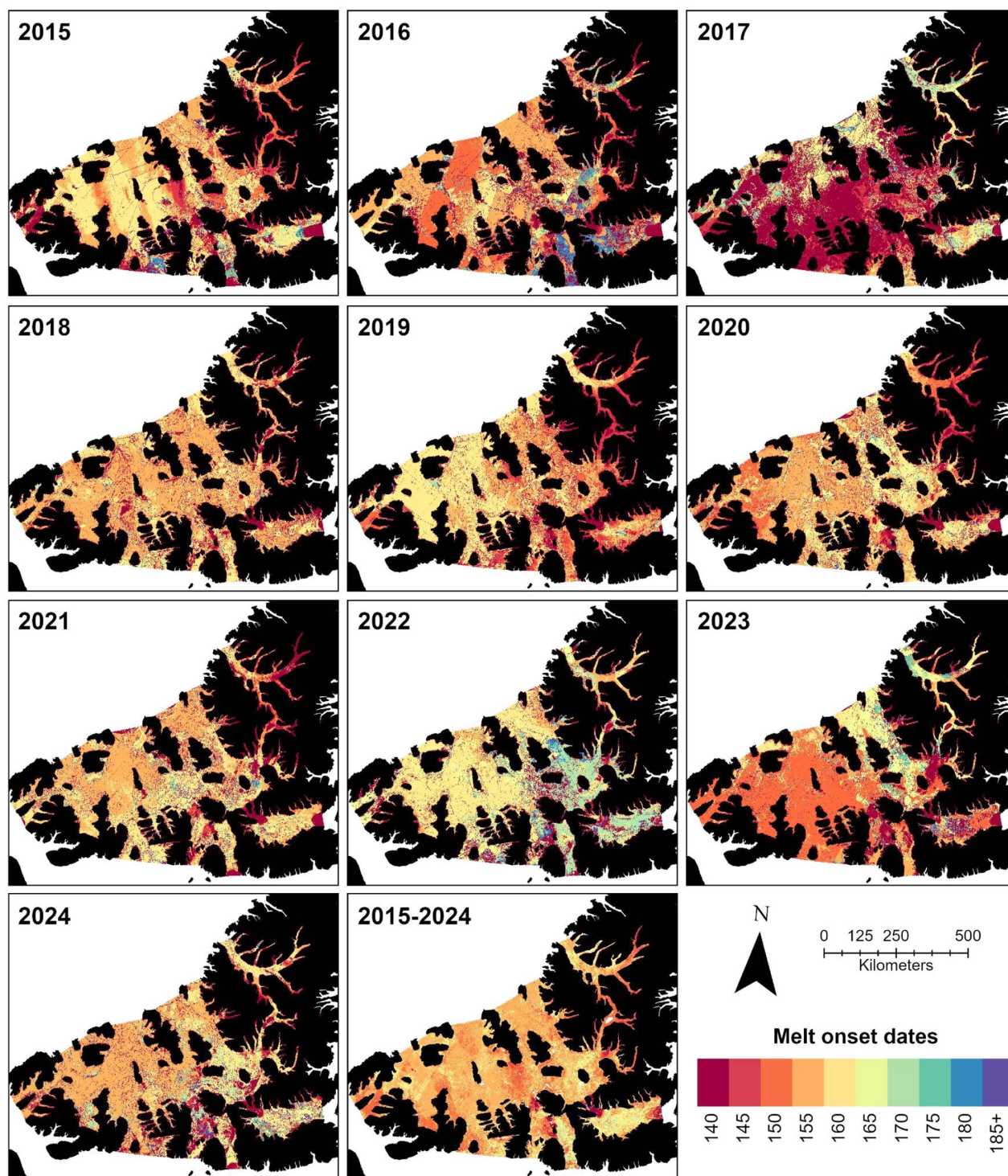


Fig. 7. Spatial distribution of the date of MO (DOY) as detected by Sentinel-1 in the QEIs for each year from 2015–2024 together with the 2015–2024 mean.

1.49 days decade⁻¹ [see Fig. 9(c)]. Using PMW observations, Howell et al. [7] reported a MO trend of -3.7 days decade⁻¹ for the QEI from 1979–2008. Also using PMW observations, Bliss et al. [9] reported the negative MO trend within the CAA between 1.7 and 3.2 from 1979–2016. Overall, this indicates that the long-term trend in PMW MO timing has become more

stable while the SAR MO long-term trend is getting stronger. The differences in trend magnitudes are reflected in methodological differences, different sensors, and slightly different study domains. However, considering the high spatial resolution of SAR imagery (~ 100 m), we would expect MO to be detected earlier than in the coarser 25 km resolution PMW

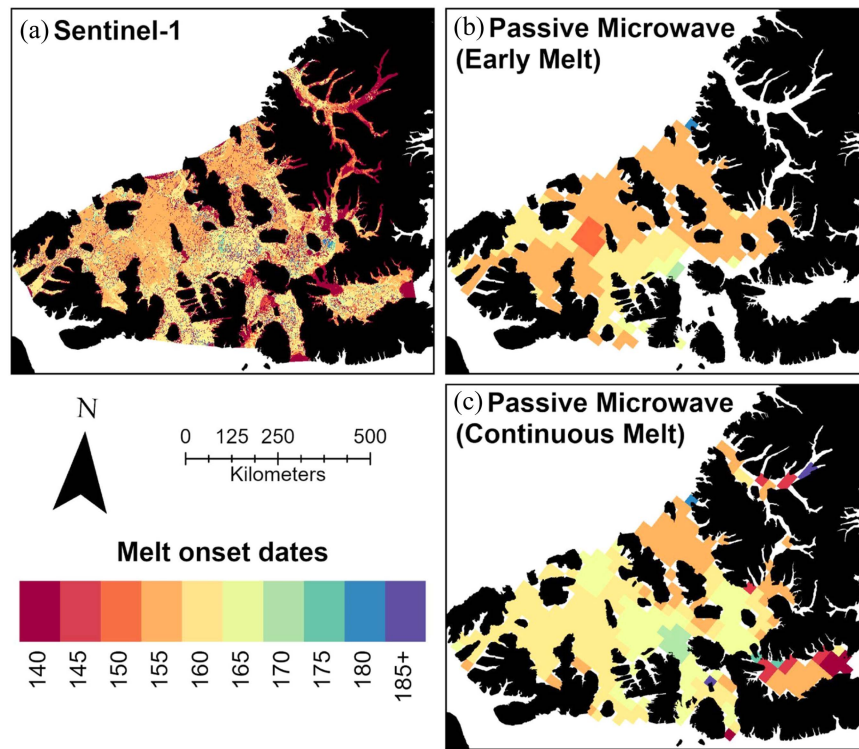


Fig. 8. Spatial distribution of the timing of MO (DOY) as detected by (a) Sentinel-1, (b) PMW early melt, and (c) PMW continuous melt in the QEIs in 2021.

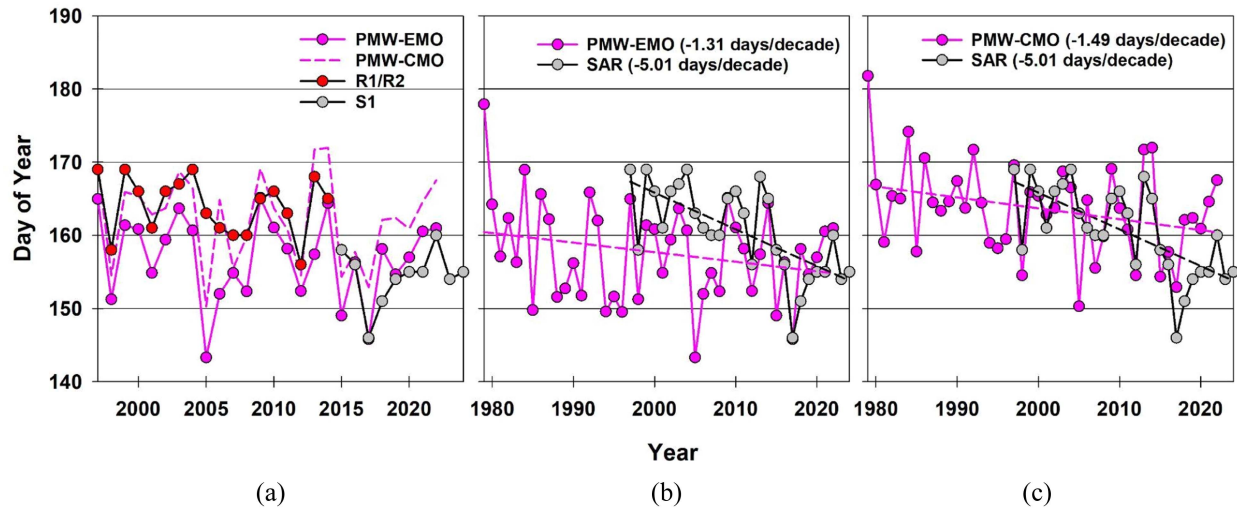


Fig. 9. (a) Time series of MO in the QEIs detected from SAR (RADARSAT-1/2 and Sentinel-1) and PMW (early and continuous melt) from 1997–2024. (b) Time series of MO in the QEIs detected by SAR from 1997–2024 and MO detected by PMW early MO from 1979–2022. (c) Time series of MO in the QEIs detected by SAR from 1997–2024 and MO detected by PMW continuous MO from 1979–2022.

observations. In other words, changes in liquid water within the snowpack sufficient to detect MO from SAR only need to occur over a much smaller region (i.e., more sensitive), compared to PMW.

In the last decade, a pronounced alteration in the lower troposphere's thermal and moisture structure in the Arctic has triggered extensive sea ice melt and will become more frequent in the future [66]. Mortin et al. [4] found that downwelling longwave radiation ($L_{\downarrow}$) was the dominant control influencing

the timing of MO. The time series of SAR MO, together with May to June (MJ) $L_{\downarrow}$ is shown in Fig. 10. The correlation between mean May to June (MJ) $L_{\downarrow}$ and SAR MO timing from 1997–2024 is statistically significant, but only -0.51 , indicating that other factors also influence the timing of MO within the northern QEI. However, there are significant divergent trends in MO timing and MJ $L_{\downarrow}$, indicating that the increases in $L_{\downarrow}$ have contributed to the observed early timing of MO in recent years (see Fig. 10).

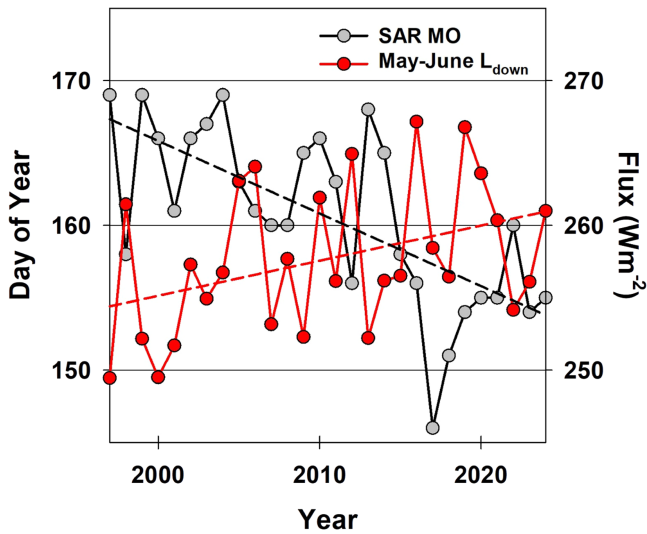


Fig. 10. Time series of SAR detected MO together with mean May to June downwelling longwave radiation ($L_{\downarrow}$) in the QEIs from 1997–2024.

V. CONCLUSION

The QEI region is located within the Last Ice Area, and changes within this region have profound implications for the overall health of Arctic sea ice. In this study, we used Sentinel-1 SAR imagery to detect the timing of MO within the QEI region of the northern CAA from 2015–2024. We also compared our results to PMW observations of MO, which have been used to discuss long-term variability and change.

Sentinel-1 SAR MO spatial maps provided considerably higher spatial detail than traditional PMW observations, especially close to the coastline or in narrow channels and inlets. Over the period of 2015–2024, the QEI region experienced considerable interannual variability in the timing of MO from SAR. $L_{\downarrow}$ was the dominant influence on MO but only experienced a correlation of -0.51 , which indicates that other factors (i.e., temperature, shortwave radiation, and snow depth) play a role. From 2015–2024, no trend toward earlier MO from SAR was found, and in fact, there was a slight increasing trend in the timing of MO from SAR that was also apparent in PMW observations.

Extending the SAR MO time series back to 1997 using RADARSAT-1/2 SAR MO observations from [15] revealed a decreasing trend of $5 \text{ days decade}^{-1}$. For the PMW record from 1979–2022, we found that MO has only decreased between 1.31 and $1.49 \text{ days decade}^{-1}$. Taken together, the SAR MO timing trend appears to be getting stronger (earlier), whereas PMW has become more stable. Indeed, differences in trend magnitudes are rooted in differences in their associated methods and input data. Still, given the higher spatial resolution of SAR imagery compared to PMW observations, we would expect the latter to be more sensitive to melt detection. It is therefore vital to maintain the same time series of MO estimates from SAR, as it provides additional insight into changes that may be muted by PMW observations.

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